

Mathematics for Social Scientists

Part 3: Linear Algebra II - Systems of Linear Equations

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Outline

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Systems of linear equations (SLEs)

A system of linear equations (SLE) with n equations and m unknowns is given by:

$$a_{11}x_1 + a_{12}x_2 + \dots + a_{1m}x_m = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \dots + a_{2m}x_m = b_2$$

$$\vdots$$

$$a_{n1}x_1 + a_{n2}x_2 + \dots + a_{nm}x_m = b_n$$

- $a_{ij} \in \mathbb{R}$ are called *coefficients*, $x_1, \dots, x_m$ *unknowns*.

This system is exactly equivalent to the matrix equation

$$Ax = b.$$

Set of solutions

The SLE problem:

Given a matrix A and vector b , find all vectors x such that $Ax = b$.

Definition 1.

The **set of solutions** $\mathbb{L}$ of an SLE is the set of all vectors $(x_1, \dots, x_m) \in \mathbb{R}^m$ such that all the n equations of the SLE are (simultaneously) fulfilled.

Remark.

Concerning the set of solutions of an SLE, the following situations can occur:

- The SLE does not have a solution: $\mathbb{L} = \emptyset$,
- the SLE has exactly one solution: $|\mathbb{L}| = 1$,
- the SLE has infinitely many solutions: $|\mathbb{L}| = \infty$.

Gauss tableau

Definition 2.

Let

$$a_{11}x_1 + a_{12}x_2 + \dots + a_{1m}x_m = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \dots + a_{2m}x_m = b_2$$

$$\vdots$$

$$a_{n1}x_1 + a_{n2}x_2 + \dots + a_{nm}x_m = b_n.$$

be a system of linear equations. The corresponding **Gauss tableau** is given by

$$\left(\begin{array}{cccc|c} a_{11} & a_{12} & \dots & a_{1m} & b_1 \\ a_{21} & a_{22} & \dots & a_{2m} & b_2 \\ \vdots & \vdots & & \vdots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nm} & b_n \end{array} \right)$$

Gauss tableau ctd

Example.

$$\begin{aligned}x_2 + 2x_3 &= 3 \\x_1 + 3x_2 - x_3 &= 5\end{aligned}$$

$\Rightarrow$

$$\left(\begin{array}{ccc|c} 0 & 1 & 2 & 3 \\ 1 & 3 & -1 & 5 \end{array} \right)$$

Elementary Row Transformations

Definition 3.

The following transformations of an SLE are called **elementary row transformations**:

- interchanging two rows of an SLE,
- addition of one row of an SLE to another row,
- multiplication of a row of an SLE with a factor $\lambda \neq 0$,
- adding a multiple of one row to another row.

(This one is just the combination of the two previous ones.)

Elementary Row Transformations ctd

Proposition 1.

Elementary row transformations do not change the set of solutions $\mathbb{L}$ of a system of linear equations.

Example.

Consider the system of linear equations:

$$\begin{aligned}x_1 + 3x_2 + 5x_3 + 3x_4 + 8x_5 &= 5 \\ -7x_1 + x_2 + 5x_3 + 6x_4 + 2x_5 &= -4 \\ 1x_2 + 3x_3 + 3x_4 - 5x_5 &= 2\end{aligned}$$

Elementary Row Transformations, eg

Example (ctd.).

The previous SLE has the following Gaussian tableau:

$$\left(\begin{array}{ccccc|c} 1 & 3 & 5 & 3 & 8 & 5 \\ -7 & 1 & 5 & 6 & 2 & -4 \\ 0 & 1 & 3 & 3 & 5 & 2 \end{array} \right),$$

which is equivalent to

$$\left(\begin{array}{ccccc|c} 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 2 & 3 & 4 & 1 \\ 0 & 1 & 1 & 0 & 1 & 1 \end{array} \right).$$

The diagonal of an SLE / Gauss tableau

Definition 4.

The **diagonal** of an SLE/Gauss tableau is given by the set of coefficients $a_{11}, a_{22}, \dots, a_{n \wedge m, n \wedge m}$ ($x \wedge y = \min(x, y)$).

Example.

$$\begin{pmatrix} 2 & 3 & -1 & 6 & -4 & 1 \\ -4 & 8 & -3 & 2 & -2 & 1 \\ 2 & 3 & -7 & 2 & -12 & 1 \end{pmatrix}$$

Row echolon form

Definition 5.

A Gauss tableau is said to be in **row echolon form** if $a_{ij} \neq 0$ only for $j \geq i$ and there is no remaining row which can be transformed to a zero row.

The Gauss tableau is said to be in **reduced row echolon form** if the leading coefficient (that is, the first non-zero coefficient in each row) is equal to 1.

This means that in row echolon form, non-zero entries are allowed only on the diagonal and to the right of it.

Example

Example.

Let's transform the following Gauss tableau into row echolon form:

$$\left(\begin{array}{ccc|c} 2 & 3 & 1 & 1 \\ 1 & 1 & 1 & -1 \\ -1 & 2 & 2 & 2 \end{array} \right)$$

Example

Example.

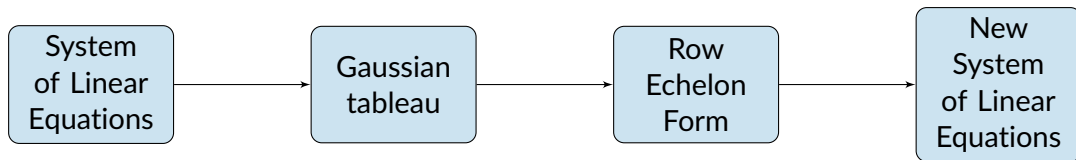
Let's transform the following Gauss tableau into row echolon form:

$$\left(\begin{array}{ccc|c} 2 & 3 & 1 & 1 \\ 1 & 1 & 1 & -1 \\ -1 & 2 & 2 & 2 \end{array} \right)$$

One possible solution:

$$\left(\begin{array}{ccc|c} 2 & 3 & 1 & 1 \\ 0 & -0.5 & 0.5 & -1.5 \\ 0 & 0 & 6 & -8 \end{array} \right)$$

Solving the SLE



The key idea: The system of linear equations obtained at the end of the process is **much simpler, but has the same solution set as the original.**

1. Solve (i.e. find the solution set) of the shortest equation.
2. Simplify the next shortest equation, and then solve in the same way.
3. Repeat until every equation in the SLE has been solved. What remains is the solution set.

From row echolon form to the solution

- Tableau in row echolon form:

$$\left(\begin{array}{ccc|c} 2 & 3 & 1 & 1 \\ 0 & -0.5 & 0.5 & -1.5 \\ 0 & 0 & 6 & -8 \end{array} \right)$$

- Back to SLE:

$$\begin{aligned} 2x_1 + 3x_2 + x_3 &= 1 \\ -0.5x_2 + 0.5x_3 &= -1.5 \\ 6x_3 &= -8 \end{aligned}$$

- Some calculations... (see whiteboard)

From row echolon form to the solution

- Tableau in row echolon form:

$$\left(\begin{array}{ccc|c} 2 & 3 & 1 & 1 \\ 0 & -0.5 & 0.5 & -1.5 \\ 0 & 0 & 6 & -8 \end{array} \right)$$

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- Some calculations... (see whiteboard)
- $\mathbb{L} = \{(-\frac{4}{3}, \frac{5}{3}, -\frac{4}{3})\}$, that is, there is exactly one solution.

Another example

Example.

-

$$x_1 + 1x_2 - 3x_3 = 1$$

$$2x_1 - x_2 - 3x_3 = 4$$

-

$$\left(\begin{array}{ccc|c} 1 & 1 & -3 & 1 \\ 2 & -1 & -3 & 4 \end{array} \right)$$

-

$$\left(\begin{array}{ccc|c} 1 & 1 & -3 & 1 \\ 0 & -3 & 3 & 2 \end{array} \right)$$

Another example

Example (ctd.).



$$x_1 + x_2 - 3x_3 = 1$$

$$-3x_2 + 3x_3 = 2$$

- Solution set

$$\mathbb{L} = \left\{ \begin{pmatrix} 4x_3 - 1 \\ 2/3 - x_3 \\ x_3 \end{pmatrix} : x_3 \in \mathbb{R} \right\}$$

The rank of an SLE

Consider a system of linear equations in **row echelon form**.

Definition 6.

- the number of rows which is not completely equal to zero without taking the right hand side of the SLE into account is called the **rank of the SLE**.
- the number of rows which is not completely equal to zero taking the right hand side of the SLE into account is called the **rank of the extended coefficient matrix**.

Example.

Following Gaussian tableau is in row echelon form.

Rank of the SLE is 2.

But the rank of the extended coefficient matrix is 3.

$$\left(\begin{array}{ccc|c} 1 & 5 & 3 & 4 \\ 0 & 2 & 3 & 1 \\ 0 & 0 & 0 & 1 \end{array} \right)$$

Connection between rank and the set of solutions

The following theorem provides a connection between rank and the set of solutions:

Theorem 2 (Rank and Set of Solutions).

Consider an SLE with n equations and m variables. Then, the following are true:

- The SLE is solvable ($\mathbb{L} \neq \emptyset$) if and only if the rank of the SLE is equal to the rank of the extended tableau/matrix.
Two subcases can occur:
 - If $\text{rank} = m$, then there is a unique solution.
 - If $\text{rank} < m$, then there are infinitely many solutions.
- The SLE is not solvable ($\mathbb{L} = \emptyset$) if and only if the rank of the SLE is strictly smaller than the rank of the extended tableau/matrix.

Some examples

- $(A, a) = \left(\begin{array}{cc|c} 1 & 5 & 4 \\ 0 & 2 & 3 \end{array} \right)$

- $(B, b) = \left(\begin{array}{ccc|c} 1 & 5 & 3 & 4 \\ 0 & 2 & 3 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right)$

- $(C, c) = \left(\begin{array}{ccc|c} 1 & 5 & 3 & 4 \\ 0 & 2 & 3 & 1 \\ 0 & 0 & 0 & 1 \end{array} \right)$

- $(D, d) = \left(\begin{array}{cccc|c} 1 & 5 & 3 & 1 & 4 \\ 0 & 2 & 3 & 1 & 1 \\ 0 & 0 & 3 & 2 & 0 \end{array} \right)$

Q: What does Theorem 2 imply about the number of solutions for the corresponding SLEs?

The case of infinitely many solutions

In the case of infinitely many solutions, we have the following further statement:

Theorem 3.

If an SLE has infinitely many solutions, then
the number of variables that can be chosen freely is equal to the difference:

$$(m - \text{rank})$$